

Finite Two-Layer Source Selection in Three-Dimensional Navier–Stokes Flow

C. K. Zeng*

9 September 2026

Abstract

I give an explicit real algebraic trigonometric-polynomial initial datum for the unforced incompressible Navier–Stokes equations on the standard three-dimensional torus at viscosity one. In the same full smooth solution, an initially empty Fourier band is generated and its actual quadratic self-interaction supplies three linearly independent target frequency pairs. On one explicit positive rational time interval, the Wiener norm of each target minus this Duhamel source is at most one seventh of the source norm; the source and target are nonzero at every positive time in that interval. The construction cancels the complete mixed four-leaf leading coefficient while retaining a nonzero balanced coefficient. Exact rational computation certifies the coefficients, both real roots of the parameter polynomial, and the original time window. A full Wiener-series construction and explicit heat-tree and infinite-tail bounds transfer those finite identities to the original equations. The result is finite two-layer source selection from one fixed datum; it asserts neither a net energy transfer nor an iterated regeneration or singularity mechanism.

2020 Mathematics Subject Classification. 35Q30, 76D05, 65G20.

Keywords. Navier–Stokes equations; Fourier–Leray interaction; Wiener algebra; fixed initial datum; exact rational certificate.

1 Introduction

A nonzero coefficient in a finite nonlinear expansion need not identify the dominant source of the corresponding mode in a solution of the full equations. Other ordered interactions can reach the same frequency, and their leading terms can be of the same size. Heat propagation and the remainder of the expansion must also be controlled. This paper constructs one explicit example in which this source comparison can be made rigorously for three target directions at once.

The datum consists of three cyclically related low-frequency pairs and three cyclically related high-frequency pairs. Each target first appears at four leaves. The complete four-leaf contribution splits into a balanced $2 + 2$ part and the sum of the $1 + 3$ and $3 + 1$ parts. A real algebraic choice of the polarizations cancels the entire mixed leading term, without cancelling the balanced term. The source studied below is nevertheless defined from the actual generated band of the same Navier–Stokes solution, not from its leading coefficient. Two separate remainder estimates establish the comparison on a common interval, including its positive endpoint.

The main contribution is this explicit fixed-datum construction and its quantitative source attribution. Local smooth existence, the heat semigroup, Wiener estimates, ordered binary trees, cyclic symmetry, and real-root isolation are used as tools. We make no claim of a new local existence principle or of priority for those tools. For the classical analytic setting see [1]; for the real algebraic algorithms used in the finite certificate see [2]. The analytic estimates

*Independent Researcher. ORCID: [0009-0002-5402-3374](https://orcid.org/0009-0002-5402-3374).

needed here are proved below, and the exact finite arithmetic is specified in Section 4 and Appendix A.

Source selection has a precise meaning in this paper: the norm of a target minus one specified Duhamel source is small relative to that source. This is not an orthogonal decomposition of kinetic energy. In particular, the factor $1/7$ is not an energy-transfer fraction. The construction concerns two finite layers; no repeated input structure or infinite-scale evolution is asserted.

Section 2 states the explicit datum and the main theorem. Section 3 proves the full-solution and source estimates for general finite data. Section 4 gives the finite coefficient and real-root certificate. Section 5 verifies the support conditions and proves the theorem for the stated datum.

2 The fixed datum and the main theorem

We use $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$, Fourier coefficients $f_k = (2\pi)^{-3} \int_{\mathbb{T}^3} f(x) e^{-ik \cdot x} dx$, and

$$\|f\|_{\mathcal{A}_m} = \sum_{k \in \mathbb{Z}^3} (1 + |k|)^m |f_k|, \quad \|f\|_{\mathcal{A}} = \|f\|_{\mathcal{A}_0}, \quad |f|_{\mathcal{A}_1} = \sum_k |k| |f_k|.$$

Vector lengths are Hermitian Euclidean lengths. A frequency dot product inside a Fourier convolution is complex bilinear. For divergence-free inputs define the full ordered bilinear operator

$$B(f, z)_k = -i\mathbb{P}_k \sum_{r+s=k} (f_r \cdot s) z_s, \quad \mathbb{P}_k = I - \frac{k \otimes k}{|k|^2} \quad (k \neq 0), \quad B(f, z)_0 = 0. \quad (1)$$

For $k = 0$ the ordered summands vanish by $f_r \cdot r = 0$. For a set $H \subset \mathbb{Z}^3$, P_H retains precisely its Fourier coefficients.

Define the rational polynomials

$$\begin{aligned} P(X) = & \frac{12379608041588}{607290075} + \frac{129169813631527}{1012150125} X + \frac{111813118362046}{337383375} X^2 \\ & + \frac{1377133612024132}{3036450375} X^3 + \frac{207263796033514}{607290075} X^4 \\ & + \frac{401161881122159}{3036450375} X^5 + \frac{4124532235492}{202430025} X^6, \end{aligned} \quad (2)$$

$$r(X) = \frac{1807613971}{12393675} + \frac{9523582624}{20656125} X + \frac{27816146846}{61968375} X^2 + \frac{196585982}{1770525} X^3, \quad (3)$$

$$s(X) = \frac{536919079}{2699067} + \frac{9469267576}{13495335} X + \frac{10524885968}{13495335} X^2 + \frac{646271818}{2699067} X^3. \quad (4)$$

Lemma 4.1 proves that P has exactly two real roots, both simple, one in each of

$$(-75/32, -125/64], \quad (-125/64, -25/16]. \quad (5)$$

Let α_* be the first root and set

$$v_* = -s(\alpha_*), \quad w_* = r(\alpha_*), \quad U = (0, 1, \alpha_*), \quad V = (3v_* + 5w_*, -2v_*, -2w_*). \quad (6)$$

There is no further scaling of (v_*, w_*) . Put $R(x_1, x_2, x_3) = (x_3, x_1, x_2)$ and $p_0 = (2, 3, 5)$, and define

$$\begin{aligned} a_*(x) = & \sum_{j=0}^2 [R^j U \{e^{i(R^j e_1) \cdot x} + e^{-i(R^j e_1) \cdot x}\} \\ & + R^j V \{e^{i(R^j p_0) \cdot x} + e^{-i(R^j p_0) \cdot x}\}]. \end{aligned} \quad (7)$$

This fixes twelve Fourier coefficients, including both signs. In particular, it fixes a single real, smooth, divergence-free, mean-zero initial datum.

Here and below $F = \text{supp } \hat{a}_*$, $L = (F + F) \setminus \{0\}$, $\kappa = (3, 4, 6)$, $D_j = \{\pm R^j \kappa\}$, and $D = D_0 \cup D_1 \cup D_2$. The three positive target vectors have determinant 91. The constants used to state the time interval are

$$\begin{aligned} \bar{A} &= \frac{13990473994187561}{829153382400}, & A_\alpha &= \frac{75}{32}, & \ell &= \frac{32000398665717304027531027}{477317255069800857600}, \\ \bar{b} &= 28\bar{A}, & \bar{E} &= 80\bar{A}^4 7^5 + 3\bar{A}\bar{b}^4, & D_f &= 2(1 + A_\alpha^2) + 2\bar{A}\bar{b}^2. \end{aligned} \quad (8)$$

Define the explicit positive rational number

$$T_* = \min \left\{ \frac{1}{2\bar{b}}, \frac{\ell}{8\bar{E}}, \frac{1}{2D_f} \right\} = \frac{\ell}{8\bar{E}}. \quad (9)$$

The equality and all rational inequalities used below are included in Lemma 4.1. This is a conservative local interval obtained from absolute remainder estimates, not a claimed time scale of amplification.

Theorem 2.1 (Finite two-layer source selection). *For viscosity one and zero external force, the datum (7) has a unique smooth Navier–Stokes solution u on $[0, T_*]$. Set $g = P_L u$ and, for $j = 0, 1, 2$, define*

$$S_j(t) = \int_0^t e^{(t-s)\Delta} P_{D_j} B(g(s), g(s)) \, ds. \quad (10)$$

Then $g(0) = 0$. For every $0 < t \leq T_*$ and every j ,

$$|g_{(1,1,0)}(t)| \geq t/2, \quad \|S_j(t)\|_{\mathcal{A}} > 0, \quad \|P_{D_j} u(t)\|_{\mathcal{A}} > 0, \quad (11)$$

$$\frac{\|P_{D_j} u(t) - S_j(t)\|_{\mathcal{A}}}{\|S_j(t)\|_{\mathcal{A}}} \leq \frac{1}{7} < \frac{1}{4}. \quad (12)$$

The same ratio bound holds for $P_D u$ and $S_D = \sum_{j=0}^2 S_j$.

The source in (10) uses the actual projection of the same complete solution at every integration time. No leading-order field or frequency projection is reset as a new initial datum. The remainder $P_{D_j} u - S_j$ retains all other contributions. The assertion concerns three frequency directions, not the rank of three velocity vectors or a repeated input profile.

3 Full solutions and quantitative four-leaf bounds

This section proves the analytic estimates used for the fixed datum. Let a be any nonzero real, divergence-free, mean-zero trigonometric polynomial, let $F = \text{supp } \hat{a}$, and put $A = \|a\|_{\mathcal{A}}$ and $K = \max_{k \in F} |k|$. In this section the viscosity is any fixed $\nu > 0$. Define the full, untruncated recursion

$$v_0(t) = e^{\nu t \Delta} a, \quad v_n(t) = \sum_{j=0}^{n-1} \int_0^t e^{\nu(t-s)\Delta} B(v_j(s), v_{n-1-j}(s)) \, ds \quad (n \geq 1). \quad (13)$$

Each v_n contains $n + 1$ leaves. In particular, if Σ_m is the set of sums of m elements of F , then

$$\text{supp } \hat{v}_n(t) \subset \Sigma_{n+1}, \quad |k| \leq (n + 1)K \quad (k \in \Sigma_{n+1}). \quad (14)$$

All coincident frequencies are added in this recursion.

Proposition 3.1 (Full-series construction). *Put $q(t) = 4AKt$ and $T_a = (8AK)^{-1}$. On $[0, T_a]$, $u = \sum_{n \geq 0} v_n$ is the unique smooth solution of the unforced Navier–Stokes equations with datum a . For every integer $m \geq 0$ and $N \geq 0$,*

$$\|v_n(t)\|_{\mathcal{A}} \leq Aq^n, \quad \|v_1(t)\|_{\mathcal{A}} \leq Aq/4, \quad (15)$$

$$\|u(t)\|_{\mathcal{A}} \leq \frac{A}{1-q}, \quad \left\| u(t) - \sum_{n=0}^N v_n(t) \right\|_{\mathcal{A}} \leq \frac{Aq^{N+1}}{1-q}, \quad (16)$$

$$\|u(t)\|_{\mathcal{A}_m} \leq \frac{A(1+K)^m m!}{(1-q)^{m+1}}, \quad (17)$$

$$\left\| u(t) - \sum_{n=0}^N v_n(t) \right\|_{\mathcal{A}_m} \leq \frac{A[1 + (N+2)K]^m m! q^{N+1}}{(1-q)^{m+1}}. \quad (18)$$

Here and below $q = q(t)$ in time-dependent estimates.

Proof. The Leray projections have norm at most one, so

$$\|B(f, z)\|_{\mathcal{A}} \leq \|f\|_{\mathcal{A}} \|z\|_{\dot{\mathcal{A}}_1}, \quad \|B(f, z)\|_{\mathcal{A}_m} \leq \|f\|_{\mathcal{A}_m} \|z\|_{\mathcal{A}_{m+1}}. \quad (19)$$

The second bound uses $1 + |r + s| \leq (1 + |r|)(1 + |s|)$. By (14),

$$\|v_n(t)\|_{\mathcal{A}} \leq K \sum_{j=0}^{n-1} (n-j) \int_0^t \|v_j(s)\|_{\mathcal{A}} \|v_{n-1-j}(s)\|_{\mathcal{A}} \, ds.$$

Let $c_0 = 1$ and $c_n = \sum_{j=0}^{n-1} c_j c_{n-1-j}$. These count plane binary trees with n internal vertices; encoding such a tree by a balanced parenthesis word of length $2n$ gives $c_n \leq 4^n$. Symmetry in j gives $\sum_j (n-j) c_j c_{n-1-j} = (n+1)c_n/2$. Since integration of s^{n-1} divides by n , and $(n+1)/(2n) \leq 1$, induction yields $\|v_n\|_{\mathcal{A}} \leq c_n A^{n+1} (Kt)^n$. The $n = 1$ estimate directly gives $A^2 Kt = Aq/4$. Thus the series and its tails satisfy (15)–(16) when $q \leq 1/2$.

The support bound also gives $\|v_n\|_{\mathcal{A}_m} \leq A[1 + (n+1)K]^m q^n$. Use $(r+1)^m \leq m! \binom{r+m}{m}$ and sum the resulting power series. For the tail set $n = N+1+r$ and use $1 + (N+2+r)K \leq [1 + (N+2)K](r+1)$. This proves (17)–(18).

In every $\mathcal{A}_m$ the Laplacian series and the double nonlinear series are absolutely and uniformly convergent: apply the preceding estimates with $m+2$ and (19) with $m+1$. Differentiating (13) and summing therefore gives $u \in C^1([0, T_a]; \mathcal{A}_m)$ and $u_t = \nu \Delta u + B(u, u)$, $u(0) = a$; at zero take the right derivative. Repeated differentiation of this equation uses only finitely many higher spatial norms and proves smoothness on the same closed interval.

For completeness, incompressibility moves the differentiated frequency from the second input to the output. Since $\sup_{x \geq 0} x e^{-\nu \tau x^2} \leq (\nu \tau)^{-1/2}$,

$$\left\| e^{\nu \tau \Delta} B(f, z) \right\|_{\mathcal{A}} \leq (\nu \tau)^{-1/2} \|f\|_{\mathcal{A}} \|z\|_{\mathcal{A}}.$$

For two continuous mild solutions with the same initial value, their difference on a subinterval of length h is bounded by its supremum norm times $2\sqrt{h/\nu}(M_u + M_u^-)$. Choosing this factor below one and covering the common compact interval by finitely many subintervals proves uniqueness. This comparison is used after existence has been established.

The recursion preserves reality, incompressibility, and zero mean. Set

$$p_k = -\frac{1}{|k|^2} \sum_{r+s=k} (k \cdot u_r)(k \cdot u_s) \quad (k \neq 0), \quad p_0 = 0. \quad (20)$$

The weighted bounds imply smoothness of this pressure and recover $u_t - \nu \Delta u + (u \cdot \nabla)u + \nabla p = 0$. $\square$

Let L and H be fixed frequency sets satisfying, throughout this interval,

$$L \cap F = \emptyset, \quad P_L v_1 = v_1, \quad P_H v_j = 0 \quad (j = 0, 1, 2). \quad (21)$$

Define $g = P_L u$ and

$$\begin{aligned} S(t) &= \int_0^t e^{\nu(t-s)\Delta} P_H B(g(s), g(s)) \, ds, \\ W(t) &= \int_0^t e^{\nu(t-s)\Delta} P_H B(v_1(s), v_1(s)) \, ds, \\ M(t) &= \int_0^t e^{\nu(t-s)\Delta} P_H \{B(v_0(s), v_2(s)) + B(v_2(s), v_0(s))\} \, ds. \end{aligned} \quad (22)$$

Thus $P_H v_3 = W + M$. Introduce the finite spatial coefficients

$$\begin{aligned} a_1 &= B(a, a), \quad a_2 = \frac{1}{2} \{B(a, a_1) + B(a_1, a)\}, \\ W_0 &= \frac{1}{3} P_H B(a_1, a_1), \quad M_0 = \frac{1}{3} P_H \{B(a, a_2) + B(a_2, a)\}. \end{aligned} \quad (23)$$

Proposition 3.2 (Actual source and complete tails). *Under (21), for $0 \leq t \leq T_a$,*

$$\|S - W\|_{\mathcal{A}} \leq C_S t^4, \quad C_S = \frac{752}{5} A^5 K^4, \quad (24)$$

$$\|P_H u - W - M\|_{\mathcal{A}} \leq C_D t^4, \quad C_D = 512 A^5 K^4, \quad (25)$$

$$\|W - t^3 W_0\|_{\mathcal{A}} \leq C_W t^4, \quad C_W = \frac{32}{3} \nu A^4 K^5, \quad (26)$$

$$\|M - t^3 M_0\|_{\mathcal{A}} \leq C_M t^4, \quad C_M = 32 \nu A^4 K^5. \quad (27)$$

In particular, if $M_0 = 0$, then

$$\|P_H u - S\|_{\mathcal{A}} \leq C_E t^4, \quad C_E = \frac{3312}{5} A^5 K^4 + 32 \nu A^4 K^5, \quad (28)$$

$$\|S - t^3 W_0\|_{\mathcal{A}} \leq C_{\text{src}} t^4, \quad C_{\text{src}} = \frac{752}{5} A^5 K^4 + \frac{32}{3} \nu A^4 K^5. \quad (29)$$

Proof. The first two conditions in (21) give $e = g - v_1 = P_L \sum_{n \geq 2} v_n$. Hence

$$\|e\|_{\mathcal{A}} \leq 2Aq^2, \quad |e|_{\mathcal{A}_1} \leq AK \frac{q^2(3-2q)}{(1-q)^2} \leq 8AKq^2, \quad |v_1|_{\mathcal{A}_1} \leq AKq/2.$$

The three ordered terms $B(e, v_1) + B(v_1, e) + B(e, e)$ have total norm at most $A^2 K(3q^3 + 16q^4)$. Integrating, with $q(s) = q(t)s/t$, yields

$$\|S - W\|_{\mathcal{A}} \leq Aq^4(3/16 + 4q/5) \leq (47/80)Aq^4 = C_S t^4.$$

The last condition in (21) and (16) give $\|P_H u - W - M\|_{\mathcal{A}} \leq Aq^4/(1-q) \leq 2Aq^4$. These are different remainders: the first includes every higher-order contribution to the actual field g .

There are five plane ordered trees with four leaves: one balanced tree for W , and four chain trees for M . For fixed leaf frequencies and admissible vertex times, the product of all leaf, intermediate, and output heat factors is $e^{-\nu \mathcal{H}}$, where $\mathcal{H} = \sum_{\text{edges } e} |k_e|^2 \Delta t_e$. At any intervening time the propagating edges partition the four leaves. If their descendant counts are m_e , then $|k_e| \leq m_e K$ and $\sum_e m_e = 4$. Consequently

$$0 \leq \mathcal{H} \leq 16K^2 t, \quad |e^{-\nu \mathcal{H}} - 1| \leq 16\nu K^2 t. \quad (30)$$

This accounts for every heat factor, including those before the first interaction.

Take absolute values before any coefficient cancellation. Each vertex costs at most the leaf count of its second subtree times K , and the sum of absolute leaf products is A^4 . The balanced tree has derivative factor $2K^3$ and time volume $t^3/3$. The four chains have factors $6K^3, 3K^3, 2K^3, K^3$ and volume $t^3/6$ each. Their absolute bounds are therefore $(2/3)A^4 K^3 t^3$ and $2A^4 K^3 t^3$, respectively. Multiplying by (30) proves (26)–(27). The fractions in (23) are precisely these vertex-time integrals with the heat factors set equal to one. Finally combine the four estimates. In (28), W cancels exactly between target and source, so C_W need not be added there. $\square$

4 Exact finite coefficient and root certification

We first regard X, v, w as formal real variables, replace α_* by X , and use $V = (3v+5w, -2v, -2w)$. Write $B = -iQ$ at every vertex. The three-vertex physical phase is $(-i)^3 = i$. At κ , define real polynomial vectors by

$$(W_0)_\kappa = iC(X; v, w), \quad (M_0)_\kappa = iZ(X; v, w), \quad Z = M_{13} + M_{31}. \quad (31)$$

The subscripts 13, 31 specify the numbers of leaves of the two root subtrees. Every vector in (31) has three components, each linear in (v, w) and of degree at most three in X . Appendix A prints all coefficients of C, M_{13}, M_{31}, Z , without deleting mixed terms. Write, for example, $C_j = C_{jv}v + C_{jw}w$.

4.1 A completely specified finite coefficient algorithm

Here is one way to reproduce the table using only rational polynomial arithmetic. Give the four labelled leaves the frequencies (e_1, e_2, e_3, p_0) and amplitudes (U, RU, R^2U, V) , with $U = (0, 1, X)$. For each of the five plane binary tree shapes

$$((ab)(cd)), \quad (a(b(cd))), \quad (a((bc)d)), \quad ((a(bc))d), \quad (((ab)c)d),$$

run all 24 permutations of these four distinct labels. A leaf returns its frequency and amplitude. A vertex whose subtrees return (r, A) and (s, B) , with m total descendant leaves, returns frequency $k = r + s$ and the polynomial vector

$$\frac{1}{m-1} \left(I - \frac{k \otimes k}{|k|^2} \right) \{ (A \cdot s)B \} \quad (k \neq 0), \quad 0 \quad (k = 0). \quad (32)$$

The dot product is bilinear. Sum the balanced shape for C , the two shapes whose first root subtree is a leaf for M_{13} , and the remaining two for M_{31} . To obtain the two columns separately use $V = (3, -2, 0)$ and $V = (5, 0, -2)$. Each complete expression is linear in that one high leaf; the common low leaves must not be added twice. The factors $1/(m-1)$ give weight $1/3$ to the balanced tree and $1/6$ to each chain.

For a target $\varepsilon R^j \kappa$, where $\varepsilon = \pm 1$, rotate all four leaf frequencies and amplitudes by R^j and multiply the frequencies, but not the real amplitudes, by ε . The same algorithm gives $\varepsilon R^j C$, $\varepsilon R^j M_{13}$, and $\varepsilon R^j M_{31}$. This also follows directly from $\mathbb{P}_{Rk} = R\mathbb{P}_k R^{-1}$ and the three differentiated vertices. After the factor i is restored, the two signs are complex conjugates. Section 5 proves that these four labelled leaves exhaust all possible contributions to the targets through four leaves in the full twelve-mode problem.

4.2 Polynomial identities and finite arithmetic conclusions

Set

$$\mathbf{M}(X) = \begin{pmatrix} Z_{1v} & Z_{1w} \\ Z_{2v} & Z_{2w} \end{pmatrix}.$$

Direct rational polynomial addition and multiplication in the printed table give

$$\det \mathbf{M} = P, \quad Z_{1v} = r, \quad Z_{1w} = s, \quad 3Y_1 + 4Y_2 + 6Y_3 = 0 \quad (33)$$

for each column Y of each of C, M_{13}, M_{31}, Z . Define the normalized polynomials

$$\mathcal{C}_j = -C_{jv}s + C_{jw}r, \quad \mathcal{Z}_j = -Z_{jv}s + Z_{jw}r. \quad (34)$$

In particular,

$$\mathcal{Z} = (0, P, -2P/3). \quad (35)$$

All objects used in the following finite calculation are thus specified by rational numbers in this paper.

Lemma 4.1 (Rational certificate). *Let $Q = P/(4124532235492/202430025)$. The following exact arithmetic statements hold.*

1. $\gcd(Q, Q') = 1$. All real roots lie in $(-25, 25)$, and the Sturm variation numbers at $-25, -75/32, -125/64, -25/16, 25$ are respectively 4, 4, 3, 2, 2. None of those endpoints is a root. Thus (5) contains exactly the two real roots of P .
2. Each of $Z_{1v}, Z_{1w}, Z_{2v}, Z_{2w}$, the three $\mathcal{C}_j$, and the three components of $(-3s + 5r, 2s, -2r)$ has greatest common divisor one with Q . Thus the original matrix has rank one at both roots, the chosen (v, w) is nonzero, and all three normalized balanced components are nonzero there.
3. On $I = [-75/32, -125/64]$, interval evaluation gives

$$\mathcal{C}_1(I) \subset \left[-\frac{82074009235302994201}{712907528601600}, -\frac{32000398665717304027531027}{954634510139601715200} \right]. \quad (36)$$

Moreover $|X| \leq A_\alpha$, $|s(X)| \leq U_v$, $|r(X)| \leq U_w$, and $|-3s(X) + 5r(X)| \leq U_{V_1}$ on I , where

$$U_v = \frac{39312542562313}{88443027456}, \quad U_w = \frac{19711525523669}{58016563200}, \quad U_{V_1} = \frac{585728660549}{472227840}. \quad (37)$$

These obey $\bar{A} = 6(1 + A_\alpha + U_{V_1} + 2U_v + 2U_w)$.

4. With the definitions (8), the three positive rationals in the minimum (9) have their unique minimum at $\ell/(8\bar{E})$. Hence $\bar{b}T_* < 1/2$, $\bar{E}T_* = \ell/8$, and $D_fT_* < 1/2$.

Exact arithmetic verification. For clarity, the complete algorithms for checking these finite assertions are given here; large intermediate quotients need not be taken as additional inputs. Represent a polynomial by its coefficients in ascending powers, trim zero trailing coefficients, and use exact rational long division. For the greatest common divisor use Euclid's successive remainders and make the last nonzero remainder monic. This gives the gcd values in items 1 and 2. In particular the monic Q is square-free; neither irreducibility nor a number-field representation is needed.

For the Sturm calculation start with (Q, Q') and append the negative remainder of the last two entries until the remainder is zero. A positive rational rescaling of any entry is allowed; a negative rescaling is not. At each indicated rational endpoint, evaluate exactly by Horner's rule, delete zero values, and count sign changes. The resulting counts are those in item 1. The Cauchy bound $1 + \max_{j < 6} |q_j| < 25$ for the monic $Q(X) = X^6 + \sum_{j < 6} q_j X^j$ places all its roots inside this interval. Sturm's theorem then gives the asserted complete real-root count and isolation; its hypotheses are square-freeness and the verified nonroot endpoints. These are the usual exact Euclidean and Sturm procedures [2].

For item 3, perform Horner evaluation in rational intervals. Addition is endpointwise; the endpoints of a product are the minimum and maximum of the four endpoint products. Begin with the leading coefficient as a singleton interval, multiply by I , and add each next coefficient. Apply this rule to $\mathcal{C}_1$, s , r , and $-3s + 5r$, constructed by (34). It yields (36) and (37); take the larger endpoint absolute value for each upper bound. Finally substitute the displayed rational constants, reduce fractions, and cross-multiply their positive denominators. The three candidate times, in their original order, are

$$\begin{array}{l} 14806310400 \\ \hline 13990473994187561' \\ 137004842759380260925278829617938421911800292198400000000 \\ \hline 41232299689611034755668076451183233780893088052675593910262741694227351226981677' \\ 181772665668062877931536384000000 \\ \hline 2738402518998898245306036855761599274683120039481' \end{array}$$

The second is strictly smaller than the others, proving item 4.

The coefficient and rational assertions in this lemma were checked by exact computation, including a separately organized four-leaf calculation and a verifier for the rational root evidence. The definitions, complete coefficient inputs, arithmetic operations, and checkable conclusions are printed here so that the argument does not require access to a particular program or private computation record. No floating-point root or time approximation is used in the proof. $\square$

5 Application to the fixed twelve-mode datum

We verify the support hypotheses for the full recursion, then apply the exact algebra and analytic bounds. These steps also prove Theorem 2.1.

5.1 All possible short-leaf target contributions

The coordinate sums of the positive low and high frequencies are 1 and 10, respectively. Modulo nine every initial frequency has coordinate sum 1 or -1 , whereas every member of $F + F$ has sum 0, 2, or -2 . Thus $L \cap F = \emptyset$. All two-leaf nonzero frequencies belong to L , and zero output vanishes by incompressibility. Therefore $P_L v_1 = v_1$ as an identity for the entire time interval, and $g(0) = 0$.

Consider a representation of κ by at most four signed initial frequencies. If h is the number of positive high leaves minus negative high leaves and l the analogous signed number of low leaves, its coordinate sum satisfies $10h + l = 13$ and $|h| + |l| \leq 4$. The only possibility is $h = 1, l = 3$. The representation must therefore use exactly one positive high leaf and three positive low leaves. The differences from the three possible high leaves are

$$\kappa - p_0 = (1, 1, 1), \quad \kappa - Rp_0 = (-2, 2, 3), \quad \kappa - R^2 p_0 = (0, -1, 4).$$

Only the first is a sum of three positive coordinate vectors. Thus the four leaves are precisely p_0, e_1, e_2, e_3 , each once. Rotation and sign reversal prove the corresponding statement for all six targets. In particular $P_{D_j} v_n = 0$ for $n = 0, 1, 2$ at all times. These support identities follow from (14), not from a test of only the initial Taylor coefficients.

It follows that the algorithm of Section 4 computes the complete four-leaf leading terms at the targets. By (35), $M_0 = 0$ on every D_j . By (36) and the restored phase, the common pair norm is

$$\omega := \|P_{D_j} W_0\|_{\mathcal{A}} = 2|\mathcal{C}(\alpha_*)| \geq \ell > 0. \quad (38)$$

Here W_0 can equivalently be defined with $H = D$ and then restricted to D_j . The factor two counts both conjugate Fourier coefficients.

5.2 Full-data bounds and the original time interval

The actual norm and support radius of (7) are

$$A = 6 \left(\sqrt{1 + \alpha_*^2} + \sqrt{(3v_* + 5w_*)^2 + 4v_*^2 + 4w_*^2} \right), \quad K = \sqrt{38} < 7.$$

The upper bounds of Lemma 4.1 give $A \leq \bar{A}$; in particular every one of the twelve modes is included. Consequently $4AK \leq \bar{b}$, and $T_* \leq (2\bar{b})^{-1} \leq (8AK)^{-1}$. Proposition 3.1 constructs the unique full smooth solution throughout $[0, T_*]$. Rotation invariance and uniqueness imply $u(Rx, t) = Ru(x, t)$, although the following bounds require only the coefficient symmetries already established.

Since

$$\bar{E} = 80\bar{A}^4 7^5 + 768\bar{A}^5 7^4,$$

it dominates both C_E and C_{src} in (28)–(29), with $\nu = 1$. For each D_j , Proposition 3.2 therefore gives

$$\|S_j - t^3 P_{D_j} W_0\|_{\mathcal{A}} \leq \bar{E} t^4, \quad \|P_{D_j} u - S_j\|_{\mathcal{A}} \leq \bar{E} t^4. \quad (39)$$

For $0 < t \leq T_*$, $\bar{E} t \leq \ell/8 \leq \omega/8$. The reverse triangle inequality gives

$$\|S_j\|_{\mathcal{A}} \geq \frac{7}{8} \omega t^3, \quad \|P_{D_j} u\|_{\mathcal{A}} \geq \frac{3}{4} \omega t^3, \quad \frac{\|P_{D_j} u - S_j\|_{\mathcal{A}}}{\|S_j\|_{\mathcal{A}}} \leq \frac{1}{7}.$$

The D_j are disjoint, so summing their Wiener norms proves the same ratio for D .

5.3 Nonzero generation of the initially empty band

Set $r_0 = (1, 1, 0)$. Its only two-leaf input pair is e_1, e_2 and its reverse. Indeed coordinate sums for two leaves force two positive low leaves. With $U = (0, 1, \alpha_*)$ and $RU = (\alpha_*, 0, 1)$, the ordered sum before projection is $(\alpha_*, \alpha_*, 1 + \alpha_*^2)$. Projection onto $r_0^\perp$ removes the first two coordinates. Both input frequencies have squared length one and the output has squared length two. Thus the complete first response is exactly

$$(v_1)_{r_0}(t) = -ite^{-2t}(1 + \alpha_*^2)e_3. \quad (40)$$

The initial coefficient at r_0 is zero. Using $1 - e^{-2t} \leq 2t$ and (16),

$$|g_{r_0}(t) + it(1 + \alpha_*^2)e_3| \leq 2(1 + A_\alpha^2)t^2 + 2\bar{A}\bar{b}^2t^2 = D_f t^2.$$

Since $D_f T_* < 1/2$, this proves $|g_{r_0}(t)| \geq t/2$ on the whole positive interval. All conclusions of Theorem 2.1 now follow.

6 Scope and verification

The conclusion identifies a specified Duhamel contribution to three target frequency pairs in one full solution. The datum is fixed before time evolves, and the source uses the actual generated band throughout the integral. Exact cancellation concerns the complete mixed leading coefficient. The mixed positive-time trees need not vanish; their heat correction is included in the fourth-order error.

The time T_* is obtained from absolute majorants and is deliberately conservative. It supplies an explicit nonempty interval and a relative source estimate, without optimizing the interval. A lower bound on a target Fourier coefficient is not a statement about the sign of the energy flux. Nor does the linear independence of the target frequencies show that their velocity polarizations form a reusable seed. The theorem does not establish repeated regeneration, an infinite cascade, norm growth, or finite-time singularity.

Verification materials. The complete rational coefficient table and the finite algorithms needed to reproduce the arithmetic conclusions are part of this manuscript. A separate local support package contains mathematical certificate data and a portable exact verifier. It is auxiliary to the printed definitions and proof, and is not a claim of formal verification in a proof assistant. The analytic estimates are proved above for the full equation, rather than inferred from a truncated numerical evolution.

AI assistance and review status. AI systems assisted the symbolic derivations, code preparation, checking, and exposition. Separate computational organizations and additional AI-assisted checks were used, but they are not independent human peer review or machine-checked formal proofs. The author is responsible for the manuscript. This version is a working manuscript and has not been independently peer reviewed.

A Complete rational coefficient table

For a row $[a_0, a_1, a_2, a_3]$, the polynomial is $a_0 + a_1X + a_2X^2 + a_3X^3$. Each pair of rows gives the coefficients of v and w in one vector component. The coefficient algorithm (32) specifies all ordered terms underlying these numbers. Together with (2)–(4), this appendix supplies all rational inputs to the arithmetic verification in Lemma 4.1.

C

Row	a_0	a_1	a_2	a_3
$1v$	$\frac{2615504}{70821}$	$\frac{2300276}{70821}$	$\frac{2615504}{70821}$	$\frac{2300276}{70821}$
$1w$	$\frac{7408984}{128527}$	$\frac{22702528}{385581}$	$\frac{7408984}{128527}$	$\frac{22702528}{385581}$
$2v$	$-\frac{53398}{2623}$	$-\frac{351272}{13115}$	$-\frac{53398}{2623}$	$-\frac{351272}{13115}$
$2w$	$\frac{618764}{128527}$	$-\frac{889852}{128527}$	$\frac{618764}{128527}$	$-\frac{889852}{128527}$
$3v$	$-\frac{346588}{70821}$	$\frac{572206}{354105}$	$-\frac{346588}{70821}$	$\frac{572206}{354105}$
$3w$	$-\frac{12351004}{385581}$	$-\frac{3190520}{128527}$	$-\frac{12351004}{385581}$	$-\frac{3190520}{128527}$

Z

Row	a_0	a_1	a_2	a_3
$1v$	$\frac{1807613971}{12393675}$	$\frac{9523582624}{20656125}$	$\frac{27816146846}{61968375}$	$\frac{196585982}{1770525}$
$1w$	$\frac{536919079}{2699067}$	$\frac{9469267576}{13495335}$	$\frac{10524885968}{13495335}$	$\frac{646271818}{2699067}$
$2v$	$-\frac{283061969}{4131225}$	$-\frac{831709144}{2950875}$	$-\frac{6974917244}{20656125}$	$-\frac{126138761}{1180350}$
$2w$	$\frac{41668527}{899689}$	$\frac{65415951}{4498445}$	$-\frac{445867592}{4498445}$	$-\frac{84482311}{1799378}$
$3v$	$-\frac{135073219}{4957470}$	$-\frac{528289184}{12393675}$	$\frac{2784071}{4131225}$	$\frac{5569154}{354105}$
$3w$	$-\frac{703593187}{5398134}$	$-\frac{973093138}{2699067}$	$-\frac{291380520}{899689}$	$-\frac{238653598}{2699067}$

M_{13}

Row	a_0	a_1	a_2	a_3
$1v$	$\frac{1862244506}{12393675}$	$\frac{29012164492}{61968375}$	$\frac{28749609676}{61968375}$	$\frac{220946972}{1770525}$
$1w$	$\frac{197843914}{899689}$	$\frac{9848113606}{13495335}$	$\frac{10737054758}{13495335}$	$\frac{221447588}{899689}$
$2v$	$-\frac{110057513}{1377075}$	$-\frac{2052627541}{6885375}$	$-\frac{780936266}{2295125}$	$-\frac{7120292}{65575}$
$2w$	$\frac{28435077}{899689}$	$\frac{12088176}{4498445}$	$-\frac{353248767}{4498445}$	$-\frac{31876662}{899689}$
$3v$	$-\frac{10831087}{495747}$	$-\frac{17522536}{495747}$	$-\frac{12718082}{2478735}$	$\frac{3538354}{354105}$
$3w$	$-\frac{117878675}{899689}$	$-\frac{989646631}{2699067}$	$-\frac{932405969}{2699067}$	$-\frac{89472686}{899689}$

M_{31}

Row	a_0	a_1	a_2	a_3
$1v$	$-\frac{10926107}{2478735}$	$-\frac{88283324}{12393675}$	$-\frac{186692566}{12393675}$	$-\frac{1624066}{118035}$
$1w$	$-\frac{56612663}{2699067}$	$-\frac{25256402}{899689}$	$-\frac{14144586}{899689}$	$-\frac{18070946}{2699067}$
$2v$	$\frac{9422114}{826245}$	$\frac{67183723}{4131225}$	$\frac{2140366}{826245}$	$\frac{405299}{236070}$
$2w$	$\frac{13233450}{899689}$	$\frac{10665555}{899689}$	$-\frac{18523765}{899689}$	$-\frac{20728987}{1799378}$
$3v$	$-\frac{8920783}{1652490}$	$-\frac{90225784}{12393675}$	$\frac{71942623}{12393675}$	$\frac{406160}{70821}$
$3w$	$\frac{3678863}{5398134}$	$\frac{5517831}{899689}$	$\frac{8323487}{385581}$	$\frac{29764460}{2699067}$

References

- [1] J. C. Robinson, J. L. Rodrigo, and W. Sadowski, *The Three-Dimensional Navier–Stokes Equations: Classical Theory*, Cambridge Studies in Advanced Mathematics 157, Cambridge University Press, Cambridge, 2016. [doi:10.1017/CBO9781139095143](https://doi.org/10.1017/CBO9781139095143).
- [2] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, second edition, Algorithms and Computation in Mathematics 10, Springer, Berlin, 2006. [doi:10.1007/3-540-33099-2](https://doi.org/10.1007/3-540-33099-2).